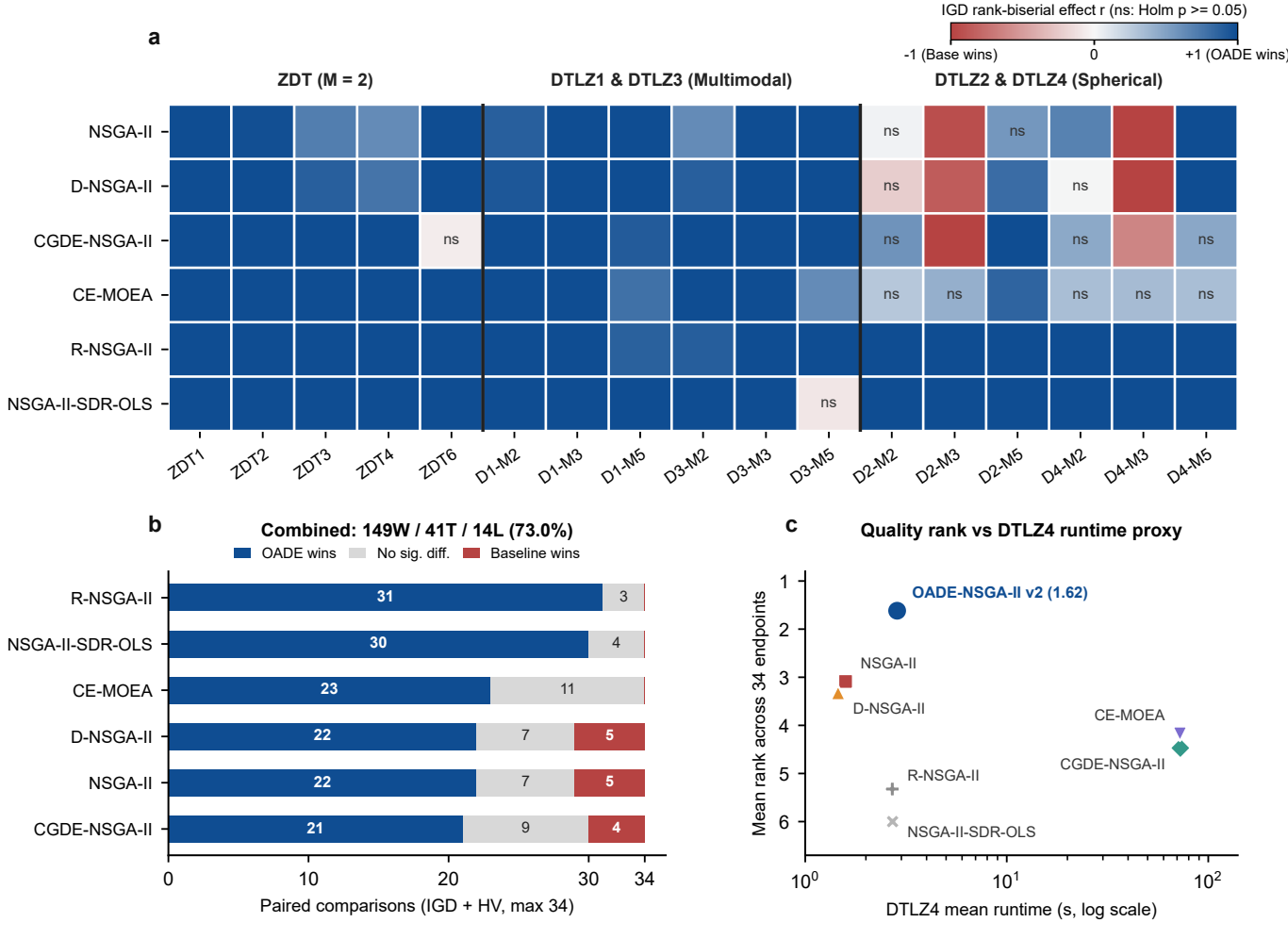

a
IGD rank-biserial effect r (ns: Holm p >= 0.05)
-1 (Base wins)
0
+1 (OADE wins)
ZDT (M = 2)
DTLZ1 & DTLZ3 (Multimodal)
DTLZ2 & DTLZ4 (Spherical)
NSGA-II
D-NSGA-II
CGDE-NSGA-II
CE-MOEA
R-NSGA-II
NSGA-II-SDR-OLS
ZDT1
ZDT2
ZDT3
ZDT4
ZDT6
D1-M2
D1-M3
D1-M5
D3-M2
D3-M3
D3-M5
D2-M2
D2-M3
D2-M5
D4-M2
D4-M3
D4-M5
ns
b
Combined: 149W / 41T / 14L (73.0%)
OADE wins
No sig. diff.
Baseline wins
R-NSGA-II
NSGA-II-SDR-OLS
CE-MOEA
D-NSGA-II
NSGA-II
CGDE-NSGA-II
31
3
30
4
23
11
22
7
5
22
7
5
21
9
4
0
10
20
30
34
Paired comparisons (IGD + HV, max 34)
c
Quality rank vs DTLZ4 runtime proxy
OADE-NSGA-II v2 (1.62)
NSGA-II
D-NSGA-II
CE-MOEA
CGDE-NSGA-II
R-NSGA-II
NSGA-II-SDR-OLS
Mean rank across 34 endpoints
1
2
3
4
5
6
10^0
10^1
10^2
DTLZ4 mean runtime (s, log scale)